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A Review on Noise Reduction in Diesel Generator Set Enclosures Using Passive Acoustic Control and Taguchi-Based Optimization Techniques

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ABSTRACT

Diesel Generator (DG) sets are extensively used as primary or standby power sources in industrial, commercial, and institutional applications. However, high noise levels generated by DG sets pose serious environmental and occupational health concerns, compelling regulatory bodies such as the Central Pollution Control Board (CPCB), India, to enforce strict noise limits. This review paper presents a comprehensive analysis of existing literature on DG set noise sources, passive noise control techniques using acoustic enclosures, and the application of Design of Experiments (DOE), particularly the Taguchi method, for optimization of enclosure parameters. The review highlights the influence of material properties such as foam density, rockwool density, and sheet metal thickness on noise attenuation performance. Existing modeling approaches, experimental strategies, and optimization techniques are critically evaluated, and key research gaps are identified to guide future investigations.

1. Introduction

Diesel Generator (DG) sets are indispensable power sources used extensively in industrial plants, commercial complexes, hospitals, airports, educational institutions, and residential facilities, primarily as standby or backup power systems [1]. In developing countries like India, where grid power reliability remains inconsistent, DG sets contribute significantly to meeting energy demands. However, despite their operational importance, DG sets are major contributors to environmental noise pollution, creating serious occupational and public health concerns [2].

Noise generated by DG sets arises mainly from combustion processes, mechanical vibrations, exhaust flow, cooling fans, and alternator operation. Typical sound pressure levels of diesel engines range between 100 and 121 dB(A), which far exceed permissible exposure limits. Prolonged exposure to such high noise levels can result in irreversible hearing loss, psychological stress, sleep disturbance, reduced work efficiency, and other physiological disorders. Recognizing

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these hazards, regulatory authorities such as the Central Pollution Control Board (CPCB), India, have mandated that noise levels from DG sets with capacities up to 1000 kVA should not exceed 75 dB(A) at a distance of one meter from the enclosure surface [3].

To comply with stringent noise regulations and growing societal demand for quieter machinery, manufacturers and researchers have focused on developing effective noise control strategies. Noise mitigation methods are broadly classified into active and passive control techniques. While active noise control is effective at low frequencies, it is complex and costly for large-scale DG set applications. Consequently, passive noise control using acoustic enclosures or canopies has emerged as the most practical, economical, and widely adopted solution [4].

An acoustic enclosure typically consists of rigid outer panels combined with internal sound-absorbing materials such as polyurethane foam and rockwool. These enclosures function by reflecting, absorbing, and dissipating acoustic energy, thereby reducing the transmission of airborne noise to the surrounding environment. However, the acoustic performance of DG set enclosures is influenced by multiple interacting parameters, including enclosure geometry, sheet metal thickness, density and thickness of absorbing materials, ventilation openings, and structural vibration characteristics. Optimizing these parameters through conventional trial-and-error methods is time-consuming, costly, and inefficient [5-11].

In this context, Design of Experiments (DOE) techniques—particularly the Taguchi method—have gained prominence for systematic analysis and optimization of enclosure design parameters. The Taguchi approach enables identification of critical factors affecting noise reduction with a minimal number of experiments while ensuring robustness against variability. Numerous studies have demonstrated the effectiveness of DOE-based optimization in reducing DG set noise levels while maintaining adequate cooling and structural integrity [12].

This review paper aims to critically examine existing research on DG set noise sources, acoustic enclosure design, sound-absorbing materials, and the application of DOE and Taguchi methods for noise optimization. By synthesizing experimental findings, modeling approaches, and identified limitations, the review highlights key research gaps and outlines future directions for developing more efficient, compliant, and sustainable DG set noise control solutions [13-16].

2. Literature Review

Diesel generator (DG) sets are major noise sources; properly designed acoustic enclosures can typically reduce noise by 8–20 dB while maintaining cooling and access.

DG noise arises from engine combustion, exhaust, cooling fan, alternator and structure-borne vibration of the frame. Typical open DGs radiate 95–108 dB(A) at 1 m, far above recommended 60–75 dB(A) limits for occupational and residential environments. CPCB India specifies 75 dB(A) at 1 m for enclosed sets, aligning with global practice.

2.1 **Passive Noise Control: Enclosure Architectures and Materials**

Basic enclosure performance

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- Simple single/double-wall boxes with mineral/glass wool, foams or composites reduce overall SPL by ~8–20 dB or 20–76% isolation, depending on rating and distance.
- Double walls with 25–50 mm gaps filled with rock/mineral wool improve insertion loss by ~10 dB over single walls.

Table 1: Representative enclosure results

Configuration (typical)	Key materials/geometry	Noise reduction	Notes
Local plywood–metal–glass wool box, 2 kW	Plywood, galvanized sheet, glass wool, cork, sponge	~9–10 dB at 2–4 m	Engine temp < 92 °C
Steel + rockwool, rectangular	Double wall, rockwool infill	~15–20 dB IL	Widely used baseline
Multilayer panel, household 2 kW	0.9 mm steel, foam, particle board, plywood	~23 dB(A) at 1 m	Shifts from “very loud” to “moderately low”
2.5 kVA enclosure, mineral wool	Steel + mineral wool (NRC 0.78)	9.5–15.5 dB from 0.6–4.2 m	Meets ~86 dB at 4 m target
GFRP double wall mini-gen	GFRP skins + rockwool	Up to 16 dB(A) at 1 m	Light, portable; forced ventilation
Portable generator plywood-foam	Plywood + foam lining	Effective broadband TL; small temp rise	(Al-Bugharbee & Jubear, 2020)

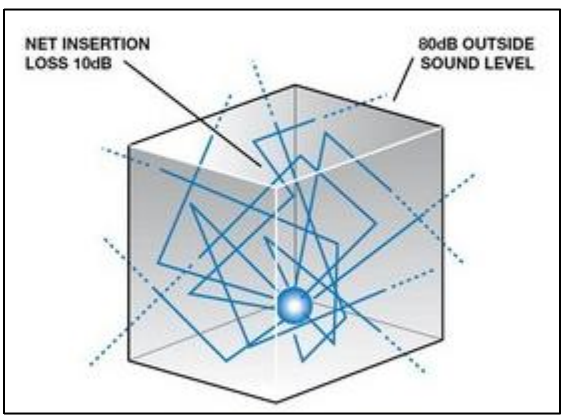

Figure 1: Typical noise reductions achieved by DG acoustic enclosures.

2. 2 Material influences and DOE/Taguchi

A Taguchi DOE on enclosure panels found rockwool density contributed more to noise reduction than foam density or steel sheet thickness, while also affecting cost. Rock/mineral wool (≈100 kg/m^3) is repeatedly selected for high absorption up to ~5 kHz, non-combustibility and good

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thermal resistance. Foam and composite panels (foam + sawdust + glass) further increase absorption but may have flammability or durability concerns.

Studies on porous foams show that density gradients and thickness optimization can tailor absorption and transmission loss, suggesting advanced graded foams or metal foams as future enclosure liners.

2.3 **Modeling, Optimization and Advanced Concepts**

Analytical and numerical modeling

- Statistical Energy Analysis (SEA) with measured source inputs predicts panel and path contributions, guiding thickness, damping and treatment placement to meet regulations before prototyping.
- Mathematical models with baffles, silencer dimensions and absorber specs generate design charts for insertion loss versus geometry, supporting preliminary sizing of enclosures and duct silencers for 50–1000 kVA gen-sets.
- Boundary element and coupled thermal–flow analyses refine enclosure geometry and ventilation duct silencers to ensure both acoustic attenuation and acceptable temperature.

2.4 **Frequency-selective and metamaterial approaches**

An adjustable acoustic "metacage" integrating reflective C-shaped unit cells with micro-perforated absorbers around an 8 kW DG achieved 13.2 and 18.4 dB IL at dominant peaks, and ~8 dB average in 100–500 Hz while preserving 25% open area for ventilation (Li et al., 2024). This demonstrates tunable low-frequency control without sacrificing airflow.

Thermal Management and Ventilation Constraints

A consistent challenge is balancing **noise attenuation with heat removal and airflow**:

- DG enclosures must pass sufficient combustion and cooling air; fully sealed designs are not feasible.
- Studies report internal generator or enclosure air temperatures remaining within safe limits (e.g., <300 °C cylinder head; ~40–92 °C enclosure air) when using ventilation openings, acoustic holes or forced-air fans.
- Duct silencers and baffled intake/exhaust paths are used to block line-of-sight sound while allowing flow, with design tuned to limit pressure drop.

Active and Hybrid Noise Control

While most DG work uses passive enclosures, active noise control (ANC) can **complement** passive measures at low frequencies (<300 Hz), where panel TL is weakest. ANC inside an enclosure using filtered-x LMS control of acoustic energy density achieved several dB reduction in global SPL, meeting a ≥2 dB(A) target around a DG. Hybrid strategies combining optimized passive treatments with ANC and improved silencers are suggested in broader generator noise reviews.

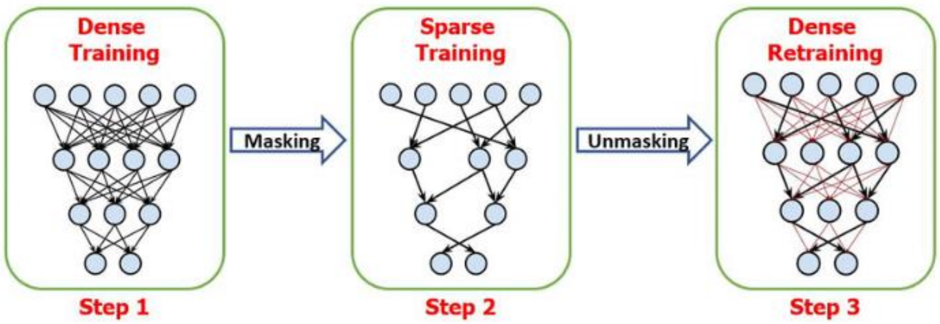

Figure 2: Areas of dense versus sparse DG noise research.

Key Research Gaps and Future Directions

- **Source characterization and path ranking**: More standardized, wide-band characterization of engine, exhaust, fan and structure-borne components across power ratings to support model-based design.
- **Material innovation**: Application of **graded foams, metal foams and metamaterials** specifically engineered for the 100–1000 Hz bands typical of DGs, while considering fire safety and cost.
- **System-level optimization**: Wider use of DOE/Taguchi, genetic algorithms and multi-objective optimization (noise, cost, weight, thermal, maintenance access) for enclosure parameter tuning .
- **Thermo–acoustic co-design**: Integrated models and experiments that jointly validate acoustic insertion loss, internal temperatures, and engine performance under real-world duty cycles.
- **Regulatory and urban context**: More work on indoor and façade-borne transmission in dense residential settings, coupling enclosure design with building acoustics and vibration isolation.

Table 2: **Research Gap Matrix**

Topic	Low-frequency (≤500 Hz) control	Wideband (≤5 kHz) control	Thermal–acoustic co-design
Conventional porous + steel/composite	4	8	4

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Topic	Low-frequency (≤500 Hz) control	Wideband (≤5 kHz) control	Thermal–acoustic co-design
Metamaterials / meta-cages	1	1	GAP
Active / hybrid control	2	1	GAP

Overall, the literature shows that well-designed passive acoustic enclosures, often using double walls with rock/mineral wool, can reliably deliver 8–20 dB noise reduction for DG sets while maintaining safe operating temperatures. Optimization tools (Taguchi DOE, SEA, design charts, genetic algorithms) are increasingly used to tune material density, thickness, panel constructions, and ventilation paths to meet regulatory limits and cost targets. Emerging metacage and hybrid active–passive approaches offer promising routes for improved low-frequency control and compact, ventilated designs, but require further experimental validation and thermo–acoustic co-optimization in realistic operating environments.

3. Major Noise Sources in Diesel Generator Sets

Diesel Generator (DG) sets produce high noise levels due to the combined effect of several mechanical, aerodynamic, combustion, and electromagnetic phenomena. Understanding and identifying the major noise sources is essential for effective noise control and optimal design of acoustic enclosures. The dominant noise sources in DG sets can be broadly classified into engine-related noise, airflow-related noise, electrical noise, exhaust noise, and structural or vibration-induced noise.

2.1 Diesel Engine Noise

Diesel engine noise is the primary contributor to overall DG set noise. It originates from rapid pressure rise during fuel combustion inside the cylinder, resulting in high-intensity impulsive forces. In addition to combustion noise, mechanical components such as pistons, connecting rods, crankshafts, valve mechanisms, and tappets generate impact and frictional noise. Phenomena such as piston slap and valve seating further intensify sound emission. Engine noise levels typically range from 100 to 121 dB(A), depending on engine size, speed, and load conditions.

2.2 Radiator and Cooling Fan Noise

Cooling systems are essential to dissipate heat generated during DG set operation; however, radiator fans are significant noise sources. High-speed rotation of the cooling fan produces aerodynamic noise due to turbulent airflow, blade-passing frequency effects, and interaction with radiator fins. The noise level from radiator fans generally ranges between 100 and 105 dB(A), increasing with fan diameter, rotational speed, and airflow resistance.

2.3 Alternator Noise

Alternator noise is generated due to cooling airflow passing through the alternator housing and mechanical interactions between brushes and slip rings. In addition, electromagnetic forces arising from magnetic flux variations in stator and rotor windings contribute to tonal noise components. Alternator noise levels usually lie in the range of 80 to 90 dB(A), but they can become prominent in enclosed configurations due to resonance effects.

2.4 Induction and Electromagnetic Noise

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Induction noise results from fluctuations in electromagnetic forces within the alternator windings. These forces cause periodic vibration of the stator core and windings, leading to audible noise emissions. Though relatively lower than engine and fan noise, induction noise becomes significant in high-capacity DG sets and contributes to overall tonal noise characteristics, typically in the range of 80 to 90 dB(A).

2.5 Exhaust Noise

Exhaust noise is produced by the release of high-pressure combustion gases from the engine cylinders into the exhaust system. This pulsating gas flow generates strong low-frequency sound waves. Exhaust silencers are commonly employed to reduce this noise, offering insertion losses of about 15–25 dB(A), while hospital-grade silencers can achieve reductions of up to 30 dB(A). Inadequate silencer design or leakage can significantly degrade overall noise control performance.

2.6 Structural and Vibration-Induced Noise

Structural noise arises from mechanical vibrations transmitted through the engine block, alternator, base frame, and mounting structures. These vibrations cause panels, enclosures, and supporting structures to radiate sound. Poor isolation, rigid mounting, or insufficient damping can amplify vibration-induced noise, leading to increased sound radiation even when airborne noise sources are controlled.

2.7 Combined Effect of Noise Sources

In practical DG set installations, the total noise level is not due to a single source but results from the combined and interacting effects of all the above noise mechanisms. Reflections within enclosures, structural resonances, and ventilation openings can further amplify noise radiation. Therefore, comprehensive noise reduction strategies must address all major noise sources simultaneously rather than focusing on individual components in isolation.

Understanding the relative contribution of each noise source forms the foundation for effective acoustic enclosure design and parameter optimization, as discussed in subsequent sections of this review

3. Application of Design of Experiments in DG Set Noise Studies

The reduction of noise in diesel generator (DG) sets involves multiple interacting design and material parameters, making conventional trial-and-error approaches inefficient, time-consuming, and costly. To overcome these limitations, Design of Experiments (DOE) has been widely adopted in DG set noise studies as a systematic and statistically robust methodology for identifying significant factors, quantifying their effects, and determining optimal parameter combinations with a minimal number of experiments.

3.1 Role of DOE in DG Set Noise Control

DOE is a structured experimental strategy that enables simultaneous variation of multiple input parameters to evaluate their individual and interactive effects on output responses such as noise level and temperature. In DG set enclosure studies, DOE facilitates:

- Identification of critical noise-influencing parameters
- Reduction in the number of experimental trials
- Development of predictive mathematical models

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- Optimization of enclosure performance under practical constraints

By applying DOE, researchers can achieve reliable and repeatable results while significantly reducing experimental cost and development time.

3.2 Commonly Investigated Parameters

Previous DG set noise studies using DOE have focused on parameters related to enclosure materials and construction. The most commonly investigated control factors include:

- Foam density (polyurethane or similar absorptive materials)
- Rockwool density and thickness
- Sheet metal thickness of the enclosure panels
- Canopy geometry and volume
- Ventilation opening size and louver design
- Engine load and operating conditions

The output responses typically analyzed are sound pressure level (dB(A)), enclosure temperature, and in some studies, enclosure weight and cost.

3.3 Taguchi Method in DG Set Noise Optimization

Among various DOE techniques, the Taguchi method is the most widely used in DG set noise reduction research. This method employs orthogonal arrays to study the influence of multiple parameters at different levels using a limited number of experiments. Taguchi's approach emphasizes robustness by minimizing variability in performance through the use of Signal-to-Noise (S/N) ratios.

For DG set noise studies, the "smaller-is-better" S/N ratio is commonly applied, as the objective is to minimize noise levels. Orthogonal arrays such as L9, L18, and L27 are frequently used depending on the number of parameters and levels considered.

3.4 Statistical Analysis and Optimization

DOE-based DG set noise studies commonly employ:

- Main effect plots to visualize parameter influence on noise level
- Analysis of Variance (ANOVA) to determine statistically significant factors and their percentage contributions
- Regression analysis to develop empirical models for noise prediction

ANOVA results in most studies indicate that material-related parameters, particularly rockwool density and foam density, contribute more significantly to noise reduction than sheet metal thickness alone. DOE also enables identification of optimal parameter combinations that achieve regulatory compliance with minimal material usage.

3.5 Advantages of DOE in DG Set Enclosure Design

The application of DOE in DG set noise research offers several advantages:

- Efficient exploration of multi-parameter design space
- Reduction in experimental cost and development time
- Improved robustness and consistency of enclosure performance
- Objective, data-driven decision-making for design optimization

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Compared to conventional single-factor experimentation, DOE provides a comprehensive understanding of parameter interactions and trade-offs.

3.6 Limitations and Challenges

Despite its advantages, DOE-based studies face certain limitations:

- Assumption of limited parameter interactions in standard orthogonal arrays
- Dependence on selected parameter ranges
- Difficulty in capturing complex nonlinear acoustic behavior

To address these challenges, recent studies have combined DOE with advanced techniques such as regression modeling, artificial neural networks (ANNs), and multi-objective optimization methods.

Overall, the application of Design of Experiments particularly the Taguchi method has proven to be an effective and practical approach for systematic investigation and optimization of DG set noise control strategies.

4. Conclusion

This review has critically examined the major noise sources in diesel generator (DG) sets, the role of acoustic enclosures in passive noise control, and the application of Design of Experiments (DOE) techniques for systematic noise reduction. DG sets remain indispensable for reliable power supply across industrial, commercial, and institutional sectors; however, their high noise emissions pose serious environmental and occupational health challenges and necessitate strict compliance with regulatory standards such as those prescribed by the Central Pollution Control Board (CPCB).

The literature clearly indicates that DG set noise is the cumulative outcome of multiple interacting sources, including engine combustion and mechanical noise, radiator and cooling fan noise, alternator and electromagnetic noise, exhaust pulsations, and structure-borne vibrations. Effective noise mitigation, therefore, requires an integrated approach rather than isolated treatment of individual noise sources. Among the various control strategies, passive noise reduction through well-designed acoustic enclosures has emerged as the most practical and widely implemented solution.

The review highlights that enclosure performance is strongly influenced by material properties such as foam density, rockwool density, and sheet metal thickness, along with enclosure geometry and ventilation design. Traditional trial-and-error approaches for optimizing these parameters are inefficient and costly. In this context, the application of DOE—particularly the Taguchi method has proven to be highly effective in identifying critical parameters, quantifying their influence, and determining optimal design combinations using a limited number of experiments.

Statistical tools such as Signal-to-Noise ratio analysis, Analysis of Variance (ANOVA), and regression modeling have been widely used to support DOE-based optimization. Most studies consistently report that sound-absorbing material characteristics contribute more significantly to noise reduction than structural thickness alone. However, the review also reveals limitations in existing research, including simplified modeling assumptions, restricted parameter ranges, and limited consideration of coupled effects such as temperature rise, vibration, enclosure weight, and cost.

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Overall, the findings suggest that DOE-based optimization offers a robust, economical, and systematic framework for improving DG set acoustic enclosure performance. Future research should focus on multi-objective optimization, advanced acoustic materials, integration of thermal and vibration analyses, and data-driven modeling techniques to develop next-generation DG set enclosures that are quieter, lighter, cost-effective, and environmentally compliant.

Conflict of Interest

The author(s) declare that there is no conflict of interest regarding the publication of this review paper.

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